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Key Points:

- An improved Empirical Canadian High Arctic Ionospheric Model driven by GNSS-TEC enhances initial value accuracy in PALSAR-CIT
- The computerized ionospheric tomography results using the IE-CHAIM model are closer to those of nearby incoherent scatter radars (ISR), that is, the true values

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GNSS-TEC Driven E-CHAIM for Enhanced PALSAR Ionospheric Tomography

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Abstract The computerized ionospheric tomography (CIT) has long been implemented in Global Navigation Satellite System systems. Recently, Phased Array L-band Synthetic Aperture Radar (PALSAR) CIT has demonstrated unique advantages, particularly achieving spatial electron density reconstruction without ground-based receivers. However, the inherent ill-posedness of tomography limits PALSAR-CIT reconstruction accuracy. To address this challenge, we calibrate the E-CHAIM by adjusting the IG parameter under the constraint of GNSS-TEC data. This integration yields a high-precision initial background field for PALSAR-CIT, effectively mitigating the ill-posedness and significantly enhancing the overall reconstruction precision. Verifications show the improved E-CHAIM (IE-CHAIM) electron density profile matches ionosonde data better than alternative methods. Simultaneously, the proposed method based on the corrected initial value is validated using three PALSAR data sets near Poker Flat, Alaska. The data measured by the incoherent scatter radar (ISR) at the corresponding time nearby are regarded as true values. The results demonstrate that introducing the IE-CHAIM as the initial background for PALSAR-CIT significantly improves the reconstruction accuracy at ISR locations compared with using the E-CHAIM, with decreases of 10.1%, 29.94%, and 26.65% in root-mean-square error (RMSE), respectively. Meanwhile, the Average Deviation (AD) values are decreased by 3.83%, 26.88%, and 35.74%, respectively. The results validate the significant role of the IE-CHAIM driven by GNSS-TEC in PALSAR-CIT. By providing optimized initial constraints, the model enables PALSAR-CIT to effectively improve the reconstruction accuracy of electron density profiles within the study area. Overall, the proposed IE-CHAIM integration acts as a critical enabling layer that transforms spaceborne SAR data into reliable 3D ionospheric insights, especially in data-sparse regions.

Plain Language Summary This paper presents a PALSAR-based computerized ionospheric tomography driven by GNSS-TEC. It extends traditional GNSS-CIT by utilizing slant TEC (STEC) derived from PALSAR data via Faraday rotation (Faraday rotation angle) estimation. The mathematical framework remains consistent with GNSS-CIT, but the signal source transitions from navigation satellites to PALSAR. First, the empirical model E-CHAIM is trained by GNSS-TEC to enhance the initial value accuracy. Subsequently, three sets of validation experiments are conducted on PALSAR-based CIT measured data, utilizing the improved initial values and the original E-CHAIM initial values as inputs, respectively. The results demonstrate that the corrected model further enhances the accuracy of the initial values, which in turn improves the final inversion accuracy of CIT. This work acts as a vital bridge to compensate for the lack of ionospheric sounding capabilities in regions where ground-based receivers are scarce, enabling reliable ionospheric monitoring.

1. Introduction

Compared with the total electron content (TEC), electron density can reflect the three-dimensional spatial variations of the ionosphere. It serves as a key physical parameter for investigating the impacts of solar activities and geomagnetic disturbances on the Earth's space environment, and directly determines the propagation characteristics of radio waves in the ionosphere. Enhancing the capability to detect ionospheric electron density (IED) is of great significance for optimizing ionospheric error correction in relevant systems, improving satellite navigation accuracy, enhancing the performance of remote sensing systems, and promoting the in-depth development of space physics research. There are various methods for measuring IED, mainly including ionosondes, radio occultation (RO) sounders, incoherent scatter radars, and in situ detectors. However, these measurement methods are usually limited to specific locations, which greatly restrict the measurement range of electron density. Aiming at this issue, Austen et al. (1988) first proposed the computerized ionospheric tomography (CIT) technique. By collecting TEC values of satellite signals along different propagation paths, CIT can reconstruct the electron density distributions in different states within a region and help understand various physical processes occurring in the ionospheric plasma. The emergence of this technique provides a solution for large-scale electron density reconstruction and improves ionospheric correction capabilities.

With the rapid development of the Global Navigation Satellite System (GNSS), GNSS-CIT research has gradually become mainstream. Regarding tomographic imaging modeling, Pryse (2003), Bust and Mitchell (2008), and Lu et al. (2021) have systematically reviewed function-based and voxel-based modeling methods both domestically and internationally, providing an in-depth discussion of their respective advantages and disadvantages. In the study of ionospheric variations, CIT has played an important role in investigating geomagnetic storm-induced ionospheric disturbances (Goswami et al., 2022; Hu et al., 2024; Kong et al., 2021; Wen et al., 2010), pre-earthquake electron density anomalies (Garcia et al., 2005; Mai & Kiang, 2009; Tang et al., 2015; Zhai et al., 2021; Wu et al., 2024), as well as electron density fluctuations caused by artificial or natural disasters such as rockets and volcanoes (Chen et al., 2024; Mei et al., 2025; Zhai et al., 2025). Concerning ionospheric correction applications, studies by Olivares-Pulido et al. (2019), and Yin et al. (2022) have shown that, compared to two-dimensional TEC, three-dimensional CIT provides more accurate ionospheric compensation information for single-station positioning, single-frequency navigation, and short-wave communication planning.

However, due to the uneven distribution of GNSS ground receivers and limitations in observational geometry, CIT suffers from a severe ill-posed problem during the inversion of the reconstruction matrix. This challenge is particularly evident in high-latitude regions, where harsh environmental conditions lead to sparse deployment of GNSS receivers, thereby limiting the reconstruction accuracy of CIT. Recently, Wang et al. (2024) conducted research on CIT using the spaceborne Phased Array L-band Synthetic Aperture Radar (PALSAR) full polarimetric (full-pol) data. Leveraging the two-way propagation characteristics of spaceborne SAR signals, this method allows the acquisition of echo scattering matrix data without the need for ground-based receiving stations. We define PALSAR-CIT as the application of CIT using TEC measurements retrieved from spaceborne PALSAR data. Methodologically, it is a logical extension of traditional GNSS-CIT; while sharing the same mathematical framework of reconstructing 3D electron density from line-of-sight observations, PALSAR-CIT uniquely overcomes the limitation of ground-based infrastructure by utilizing the two-way propagation of radar signals. This study further refines the technique into a two-stage process: 'background field construction' via model optimization and 'actual inversion' using satellite observations. From these data, the TEC along the propagation path can be retrieved and subsequently used as input for reconstruction. The results demonstrate that the spatial distribution of electron density can be reconstructed using the PALSAR data alone, providing a novel approach for IED sounding. However, the reconstruction accuracy remains highly sensitive to the selection of initial values. When high-precision initial values are used, the reconstruction results better reflect the actual distribution; otherwise, significant deviations occur, particularly in peak electron density and altitude (Prol et al., 2017; Long et al., 2023). Initial values are typically obtained from empirical models such as NeQuick2, IRI and E-CHAIM, which can characterize the average state of the ionosphere but often fail to capture short-term variations effectively. To address this issue, some previous studies adopt a model correction strategy inspired by related techniques as shown in Nava et al. (2011), Ssessanga et al. (2015), and Aa et al. (2018), in which a threshold is set (e.g. 3 TECU), and the root-mean-square error (RMSE) between the model-derived and measured TEC data (from GNSS or GIM) is calculated. If the RMSE exceeds the threshold, the core input parameter of solar or ionosphere in the empirical model (NeQuick or IRI) is adjusted to optimize the model until the RMSE falls below the threshold. The corrected model output is then used as the initial value for ionospheric tomography. Due to the fact that this research area is located in the high latitude region of the Northern Hemisphere, similar to the refinement method of the empirical model mentioned above, the Empirical E-CHAIM model is driven by GNSS-TEC to provide higher precision iteration initial values. This method reduces the discrepancy between the model and actual observations, thereby improving the accuracy of PALSAR-CIT. In this study, the electron density profiles from two ionosonde stations are utilized to verify the precision of the initial values, and three sets of PALSAR full-pol observations over the Alaska region are used for ionospheric tomography. Furthermore, the Incoherent

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Scatter Radar (ISR) nearby serve as truth to verify that the proposed method can further enhance the performance of PALSAR-CIT.

The GNSS-driven E-CHAIM bridges the gap between empirical modeling and CIT. By mitigating the ill-posedness in receiver-sparse polar regions, this framework provides a vital pathway for independent 3D reconstruction. Ultimately, this regional solution enables reliable ionospheric CIT using spaceborne SAR systems. In addition, high-latitude regions are prioritized because PALSAR's fixed 23° incidence angle yields $3 \sim 4$ times higher Faraday Rotation (Faraday rotation angle [FRA]) sensitivity there than at mid-to-low latitudes, where weak FRA signals are often obscured by system noise (Kim, 2013). This physical constraint makes the high-latitude regions the most feasible domain for verifying the proposed PALSAR-based CIT framework. Consequently, the primary focus of this paper is dedicated to high-latitude regions to effectively demonstrate the feasibility and correctness of the methodology.

The outline of this paper is given as follows: In Section 2, the basics of the E-CHAIM model driven by GNSS-TEC, the TEC retrieval based on full-pol PALSAR, and the tomography imaging algorithm are introduced in this paper. In Section 3, using three sets of full-pol PALSAR measured data in Alaska, the results are evaluated to show the feasibility of the proposed algorithm. Finally, our conclusions are put in Section 4.

2. Methodology

2.1. E-CHAIM Model Driven by GNSS-TEC

E-CHAIM is designed as an empirical electron density model for the high latitude ionosphere, aimed at replacing IRI (International Reference Ionosphere) data for that latitude region. This model can output the distribution of IED in the region above 50°N north magnetic latitude and consists of multiple sub models, each corresponding to a key feature of the IED profile. Such as the seasonal variability is modeled by a Fourier expansion and solar cycle variability is modeled via a function of solar F10.7 cm flux and IG ionospheric index. The model parameterizations for the F2 peak and topside can refer to published research (Themens et al., 2018, 2019). The output results of the ECHAIM model mainly depend on the solar (F10.7, F10.7_81, and R12) and Ionosonde Global (IG) monthly indices, among which The IG index is the Ionosonde Global monthly index based on noontime foF2 ionosonde measurements recorded at four reference stations: two from the Southern Hemisphere (Port Stanley/UK and Canberra/Canada) and two from the Northern Hemisphere (Kokubunji/Japan and Chilton/UK). The index takes into account both solar activity and ionospheric information, which can better reflect the variation of the ionosphere (Liu et al., 1983). Although some previous studies have mostly used external Vertical Total Election Content (VTEC) data to adjust F10.7 to drive more accurate electron density, it has been proven that IG is still outperforms R12 and F10.7 indices in describing solar cycle variations in the F region ionosphere (Bilitza, 2001; Bilitza et al., 2017), simultaneously, since F10.7 is fundamentally a proxy for solar radiation flux, it typically governs the computation of other coupled physical parameters within ionospheric models. Consequently, substantial adjustments to F10.7 may induce non-physical changes in the shape of the electron density profile. Recent studies [for example, Pignalberi et al., 2024] point out that the F2-layer peak serves as the anchor point for the electron density profile. Adjusting the IG index effectively constrains this anchor point, thereby successfully updating and enhancing the accuracy of the electron density profile. In addition, many previous studies have also shown that using the IG index to refine the empirical ionosphere model greatly improves the accuracy of its output electron density (Liu et al., 2019; Yao et al., 2020; Peng et al., 2024). Therefore, we adjusted the IG index to drive the ECHAIM model in this study.

Due to the CIT typically utilizes about 15-min intervals, the temporal resolution of the external VTEC data used to drive the E-CHAIM model should align with this timeframe; therefore, we use ionospheric products (VTEC) of Madrigal Database provided by the Massachusetts Institute of Technology (MIT) Haystack Observatory to drive E-CHAIM. These VTEC products offer a high spatiotemporal resolution of $1° \times 1° \times 5$ min. The MIT Haystack Observatory is a research institution dedicated to the study of atmospheric science, ionospheric physics, radio astronomy and related technologies. The Madrigal Database collects and organizes data from multiple ionospheric observation stations around the world, including radar, satellite, ground observations and other data sources. Figure 1 illustrates the global VTEC distribution obtained by fusing the Madrigal VTEC product data within a 15-min window. As observed, the fused VTEC data achieves complete spatial coverage in the study region, and its temporal resolution remains consistent with the tomographic requirements. Furthermore, it should be noted that if the observation epoch falls within this 15-min window, the VTEC data within this window can

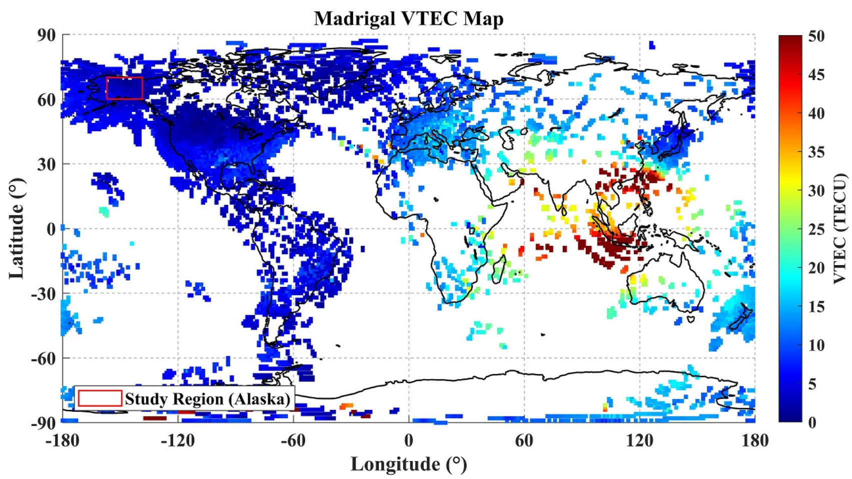

Figure 1. Distribution of Madrigal GNSS-VTEC and Alaska region.

serve as a reasonable approximation for that specific instant. Expanding this time window might degrade the data accuracy. Therefore, it is essential to ensure the completeness of the external VTEC data prior to the model refinement.

As illustrated in Figure 2, Similar to previous research methods (Aa et al., 2022; Zhang et al., 2021; Peng et al., 2024), the procedure consists of the following steps. Step 1: We utilize the initial IG parameters from the E-CHAIM database to calculate the VTEC at the grid points. Step 2: We compute the root mean square error (RMSE) for the grid points by comparing the calculated VTEC with the external VTEC data within a 15-min window. Step 3: We evaluate whether the RMSE has reached its minimum value. Step 4: If the RMSE is not minimized, the IG index is iteratively adjusted until the optimal updated index (IGup) that minimizes the local RMSE is identified. Step 5: Finally, the obtained IGup is fed back into the E-CHAIM model to compute the optimal electron density. The RMSE is defined as

$$\mathrm{RMSE} = \sqrt{\sum_{i=1}^{K} (\mathrm{VTEC}_{E-\mathrm{CHAIM}}(\mathrm{IG}) - \mathrm{VTEC}_{\mathrm{Madrigal}})/K} \tag{1}$$

where K is the number of available VTEC measurements at each grid.

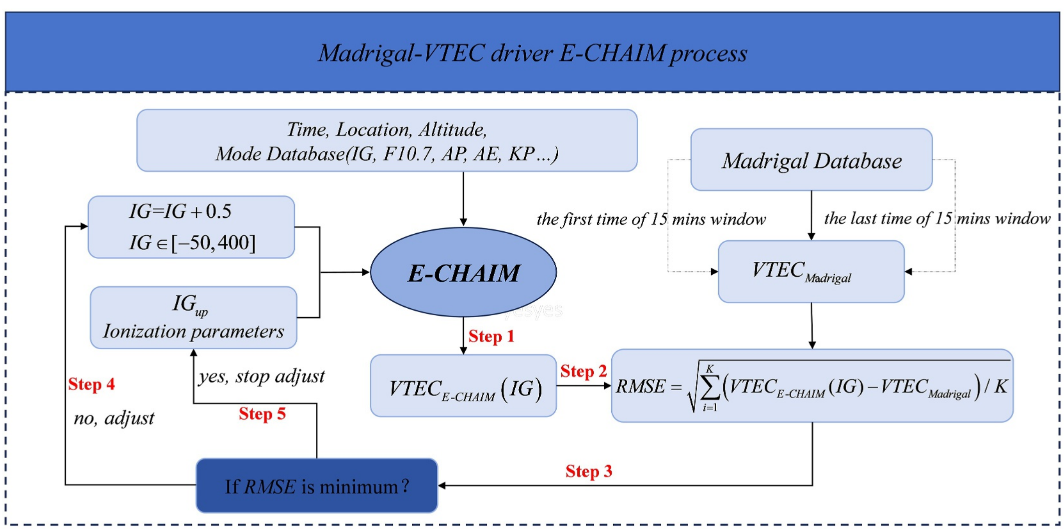

Figure 2. Process of the GNSS-TEC driven E-CHAIM model.

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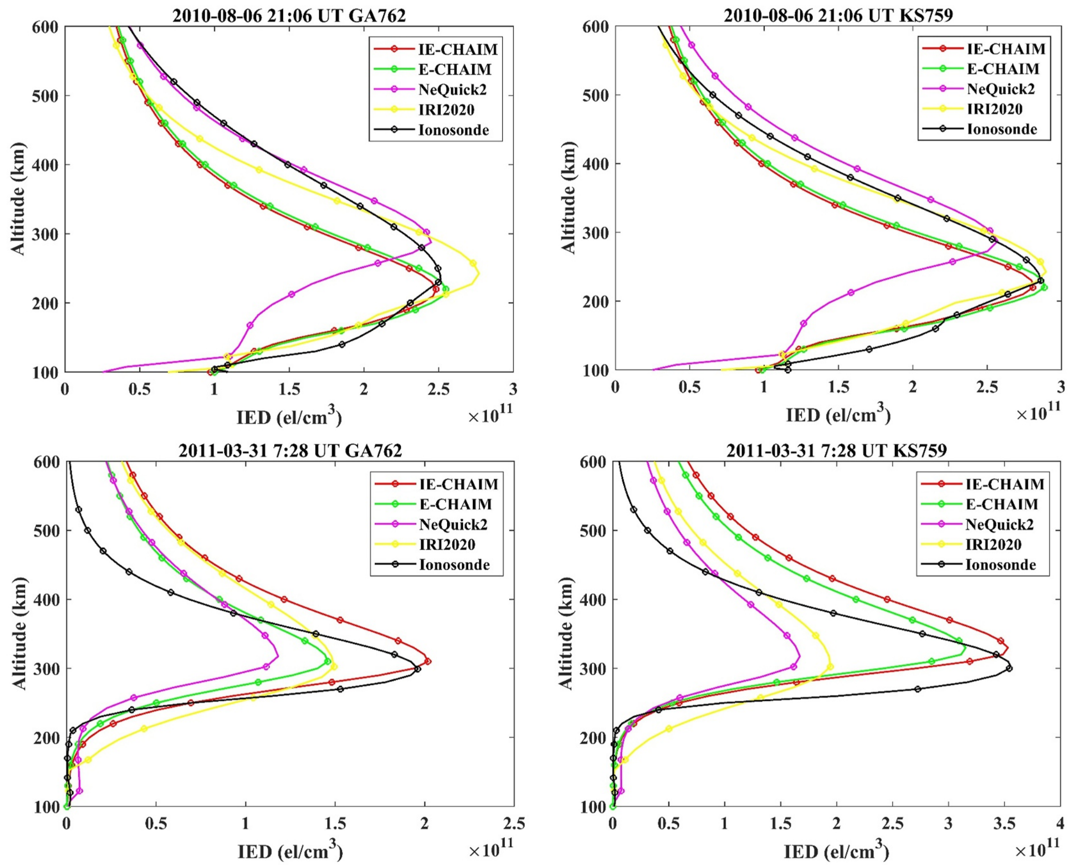

Figure 3. Comparison of ionospheric electron density (IED) profiles obtained using four methods with IED profiles from the GA762 (left) and KS759 (right) ionosonde stations during two epochs.

To verify the performance of the improved E-CHAIM (IE-CHAIM), we use two ionosonde stations GA762 (62.4°N, −145°W) and KS759 (58.4°N, −156.4°W) near the study area for accuracy verification at two corresponding epochs, where PALSAR data at three epochs (21:06 UT on 6 August 2010, 7:30 UT on 19 March 2011 and 7:28 UT on 31 March 2011) are used for CIT in the following subsection, but the ionosonde data for the second epoch is missing. Figure 3 shows the comparison between the IED profiles obtained using these four different methods (the IE-CHAIM, E-CHAIM, NeQuick2 and IRI2020) and the ionosonde station. It can be seen that the IED profiles of the improved IE-CHAIM have higher accuracy and are smoother than other methods. In addition, the electron density above the peak height of the ionosonde station is fitted using the Chapman function; therefore, the peak electron density of the ionospheric F2-layer (NmF2) is also considered an important parameter for evaluating inversion accuracy. Figure 4 shows the comparison of NmF2 from four models and ionosonde at five different periods of 3 days, while the observation data from the GA762 ionosonde station is severely missing over the second day. It can be seen from Figure 4 that the peak values of the IE-CHAIM are more in line with the ionosonde observations in almost all periods.

To further verify the accuracy of the IE-CHAIM, the root means square error ($\mathrm{RMSE_{NmF2}}$) and absolute error ($\Delta \mathrm{E_{NmF2}}$) between the NmF2 obtained by four models and ionosonde are used for comparative analysis,

$$\begin{aligned} \mathrm{RMSE_{NmF2}} &= \sqrt{\frac{1}{q}\sum_{i=1}^{q}\left(\mathrm{NmF2}_i^{\mathrm{Mod}} - \mathrm{NmF2}_i^{\mathrm{Ion}}\right)^2}, \\ \Delta \mathrm{E_{NmF2}} &= \frac{1}{q}\sum_{i=1}^{q}\left|\mathrm{NmF2}_i^{\mathrm{Mod}} - \mathrm{NmF2}_i^{\mathrm{Ion}}\right| \end{aligned} \tag{2}$$

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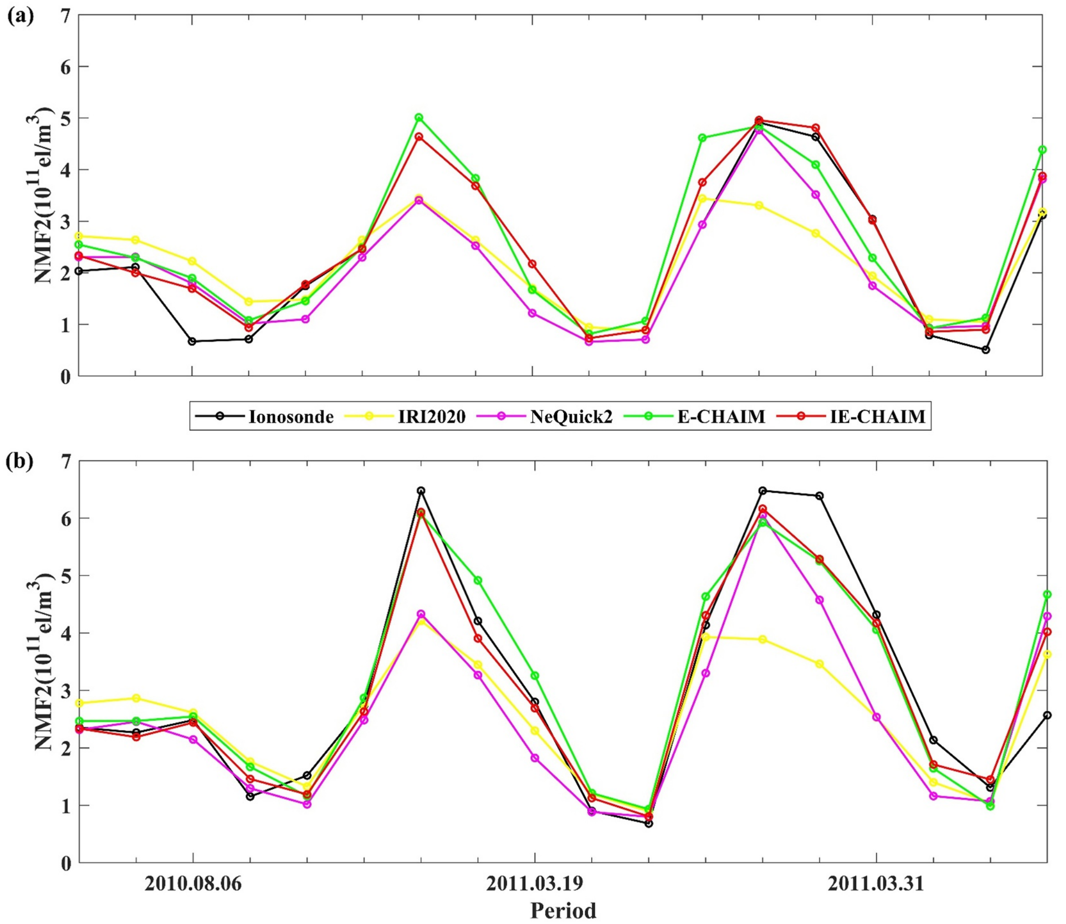

Figure 4. Comparison of peak values between ionospheric electron density estimated from four models and ionosonde data on five period of 3 days. Subfigures denote different ionosonde stations: (a) GA762 and (b) KS759.

where q is the dimension of test sample, $NmF2_i^{Mod}$ is obtained by the four models, and $NmF2_i^{Ion}$ is the observed truth value obtained from ionosonde data. It can be seen from Table 1 that the results of IE-CHAIM methods have lower errors than E-CHAIM in all periods, and the average RMSE of IRI2020, Nequick2, E-CHAIM, and IE-CHAIM are about 0.97, 0.76, 0.66, and 0.46, respectively, and the average are about 0.91, 0.62, 0.52, and 0.33, respectively. Overall, the accuracy of the initial electron density values has been significantly improved during the corresponding periods of 3 days for the following PALSAR-CIT.

Table 1
Comparison of the Errors (RMSE/ ΔE) Between the Peak Value of Four Models and Ionosonde Stations (Unit: 10^{11} El/m^3)

Ionosonde	Time	IRI2020	Nequick2	E-CHAIM	IE-CHAIM
GA762	6 August 2010	0.613/0.783	0.506/0.379	0.578/0.433	0.445/0.280
	19 March 2011	/	/	/	/
	31 March 2011	0.901/1.080	0.702/0.592	0.762/0.614	0.438/0.292
KS759	6 August 2010	0.470/0.401	0.240/0.215	0.277/0.227	0.191/0.147
	19 March 2011	1.026/0.744	1.041/0.728	0.648/0.550	0.439/0.341
	31 March 2011	1.830/1.553	1.327/1.161	1.032/0.788	0.781/0.593

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2.2. Extracted TEC Based on Full-Pol PALSAR

Due to insufficient system isolation and the complexity of electromagnetic wave propagation conditions, full-pol PALSAR measured data often exhibit distortions. The scattering matrix is used to describe the backscattering characteristics of targets and therefore must undergo calibration to ensure accurate inversion of physical parameters (Meyer & Nicoll, 2008). Considering the residual system errors after calibration, the actual scattering matrix M measured by SAR can be expressed as

$$\begin{bmatrix} M_{\mathrm{hh}} & M_{\mathrm{vh}} \\ M_{\mathrm{hv}} & M_{\mathrm{vv}} \end{bmatrix} = \begin{bmatrix} 1 & \delta_2 \\ \delta_1 & f_1 \end{bmatrix} \cdot \begin{bmatrix} \cos\Omega & \sin\Omega \\ -\sin\Omega & \cos\Omega \end{bmatrix} \cdot \begin{bmatrix} S_{\mathrm{hh}} & S_{\mathrm{vh}} \\ S_{\mathrm{hv}} & S_{\mathrm{vv}} \end{bmatrix} \cdot \begin{bmatrix} \cos\Omega & \sin\Omega \\ -\sin\Omega & \cos\Omega \end{bmatrix} \cdot \begin{bmatrix} 1 & \delta_3 \\ \delta_4 & f_2 \end{bmatrix} + \begin{bmatrix} N_{\mathrm{hh}} & N_{\mathrm{vh}} \\ N_{\mathrm{hv}} & N_{\mathrm{vv}} \end{bmatrix}, \quad (3)$$

where δ_i and f_i terms represent the cross-talk and imbalance factors, respectively. The parameter Ω denotes the FRA of linearly polarized waves caused by ionosphere and the Earth's magnetic field. S denotes the actual target scattering matrix containing the physical backscattering information from the observed scene. The matrix of N denotes the accounts for additive system noise.

Assuming that the radar signals transmitted by PALSAR are affected by the ionosphere, and that the system errors have been well calibrated and can be neglected, the relationship between the received scattering matrix M and the actual scattering matrix S can be expressed as (Li et al., 2014)

$$\begin{bmatrix} M_{\mathrm{hh}} & M_{\mathrm{vh}} \\ M_{\mathrm{hv}} & M_{\mathrm{vv}} \end{bmatrix} = \begin{bmatrix} \cos\Omega & \sin\Omega \\ -\sin\Omega & \cos\Omega \end{bmatrix} \cdot \begin{bmatrix} S_{\mathrm{hh}} & S_{\mathrm{vh}} \\ S_{\mathrm{hv}} & S_{\mathrm{vv}} \end{bmatrix} \cdot \begin{bmatrix} \cos\Omega & \sin\Omega \\ -\sin\Omega & \cos\Omega \end{bmatrix}. \quad (4)$$

Driven by the need for accurate estimation of the FRA based on Equation 3, numerous estimators have been developed (Bickel & Bates, 1965; Li et al., 2014, Chen & Quegan, 2010; Freeman, 2004). In this work, we utilize the estimator introduced by Bickel and Bates (1965), which operates on the scattering matrix in the circular polarization basis and is widely regarded for its robustness and suitability in full-polarimetric SAR-based FRA retrieval (Ji et al., 2025). Thus, the representation of Equation 3 in the circular polarization basis can be written as

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} S_{\mathrm{hh}} - S_{\mathrm{vv}} + 2jS_{\mathrm{hv}} & (S_{\mathrm{hh}} + S_{\mathrm{vv}})\, e^{\left(j\frac{\pi}{2} - j2\Omega\right)} \\ (S_{\mathrm{hh}} + S_{\mathrm{vv}}) \cdot e^{\left(j\frac{\pi}{2} + j2\Omega\right)} & S_{\mathrm{vv}} - S_{\mathrm{hh}} + 2jS_{\mathrm{hv}} \end{bmatrix}. \quad (5)$$

$\Omega_{B\&B}$ can then be derived by

$$\Omega_{B\&B} = -\frac{1}{4} \arg\left(\overline{Z_{12} \cdot Z_{21}^{*}}\right), \quad (6)$$

where $\bar{\cdot}$ and $(\,)^*$ denotes averaging and conjugate, respectively. The slant TEC can therefore be derived as

$$\mathrm{TEC}_{\mathrm{retri}} \approx \frac{\Omega_{B\&B} \cdot f_0^2}{2.365 \times 10^4 B_0 \cos\theta_B}, \quad (7)$$

where f_0 is the carrier frequency, B_0 is the magnitude of ambient magnetic field, and θ_B is the angle between the propagation direction of the radar signal and the geomagnetic field.

2.3. PALSAR-CIT Based on the Corrected Initial Values

Similar to traditional GNSS-CIT research, PALSAR-CIT also encounters the ill-posed problem arising from the limited number of satellites and sparse signal rays. The slant TEC over a special signal path l is described as

$$\mathrm{TEC} = \int_l \mathrm{Ne}(\mathbf{r})\, ds. \quad (8)$$

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Essentially, the core of CIT lies in utilizing slant TEC observations along the propagation path from multiple viewing angles, and employing retrieve algorithms to reconstruct the IED function $\mathrm{Ne}(\mathbf{r})$ in the spatial domain, where $\mathbf{r}$ is the position vector composed of longitude, latitude, and altitude.

In order to solve this problem, the voxel-based CIT model is used in this paper (Rius et al., 1997). First, the reconstructed ionosphere can be discretized into J three-dimensional voxels along the directions of longitude, latitude, and altitude, assuming that the electron density within each grid is uniform. As discussed in the above subsection, when the full-pol PALSAR observes the Earth, it can simultaneously acquire M TEC values over the observed region. Each TEC value along a ray path can be considered as the sum of the products of the electron density within the intersected voxels and the corresponding path lengths through those voxels. Therefore, the TEC value of the ith ray can be expressed as based

$$\mathrm{TEC}_i = \sum_{j=1}^{J} A_{\mathrm{ij}} x_j + \varepsilon_i, \tag{9}$$

where A_{ij} represents the path length of the ith ray within the jth voxel, x_j is the electron density in the jth voxel, and ε_i is the error associated with the ith ray. The CIT problem can thus be formulated as solving the following linear equation, which allows for the determination of the electron density distribution in each voxel, specifically

$$\mathbf{y}_{M\times 1} = \mathbf{A}_{M\times J}\mathbf{x}_{J\times 1} + \mathbf{e}_{M\times 1}, \tag{10}$$

where $\mathbf{y}$ denotes the $M \times 1$ retrieve vector of the PALSAR, $\mathbf{A}$ represents the $M \times J$ matrix formed by the intercepts of rays within the corresponding voxel, and $\mathbf{e}$ is the error vector corresponding to $\mathbf{y}$.

In general, the matrix is often not squared, ill-posed and ill-conditioned due to the sparse PALSAR radar signals within the reconstruction area, and restricted observation angles. Therefore, many methods have been employed to overcome the problem of ill-posedness in GNSS tomographic equation, such as iterative algorithms based on Algebraic Reconstruction Technique (ART), multiplicative ART (MART), simultaneous iterative reconstruction technique (SIRT), regularization method, Kalman filtering and parameterized approach (Austen et al., 1988; Raymund et al., 1990; Pryse et al., 1993; Kunitsyn et al., 2011; Song et al., 2021; Chen et al., 2019), and non-iterative algorithms based on Singular Value Decomposition (Bhuyan et al., 2002). Among the various methods, MART has been proven to provide optimal results with minimal processing time due to its advantage of avoiding time-consuming matrix operations (Bender et al., 2011). The MART algorithm gives an initial value of IED to each voxel using the improved E-CHAIM model before iteration and then gradually improves the accuracy of IED within the reconstruction area through an iterative method (Bust & Mitchell, 2008; Raymund et al., 1990). The fundamental iterative equation is

$$\mathbf{X}_j^{(k+1)} = \mathbf{X}_j^{k}\left(\frac{y_i}{\langle X^k, a_i\rangle}\right)^{\frac{\lambda_k a_{\mathrm{ij}}}{\|a_i\|}}, \tag{11}$$

where $\mathbf{X}_j^{(k+1)}$ is the IED of the jth voxel at $k+1$st iteration, and λ_k is the relaxation factor of each iteration, with a value range of $0 < \lambda_k < 1$. $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$ denote the inner product and the norm, respectively.

Thus, a critical limitation of the voxel-based approach becomes evident, namely, the scarcity or uneven spatial distribution of available observational data, which is particularly problematic for PALSAR. As a result, after discretizing the ionospheric region for inversion, some pixels may not be traversed (i.e., modulated) by any signal ray paths, leading to a strong dependence of the reconstruction results on the choice of initial values. Fortunately, one of the core tasks of this paper is to improve the traditional E-CHAIM model and apply it to the initial values of PALSAR-CIT, thereby further enhancing the authenticity of the initial values. The relevant detailed discussion and verification are presented in Section 2.1. The main purpose of this subsection is to propose a scheme for conducting PALSAR-CIT using the modified initial values, which will be verified in the subsequent section, so as to highlight the importance of the initial values. Figure 5 shows the flowchart of CIT using the improved E-CHAIM, which includes 6 steps. The first four steps pertain to the parameter configuration and spatial

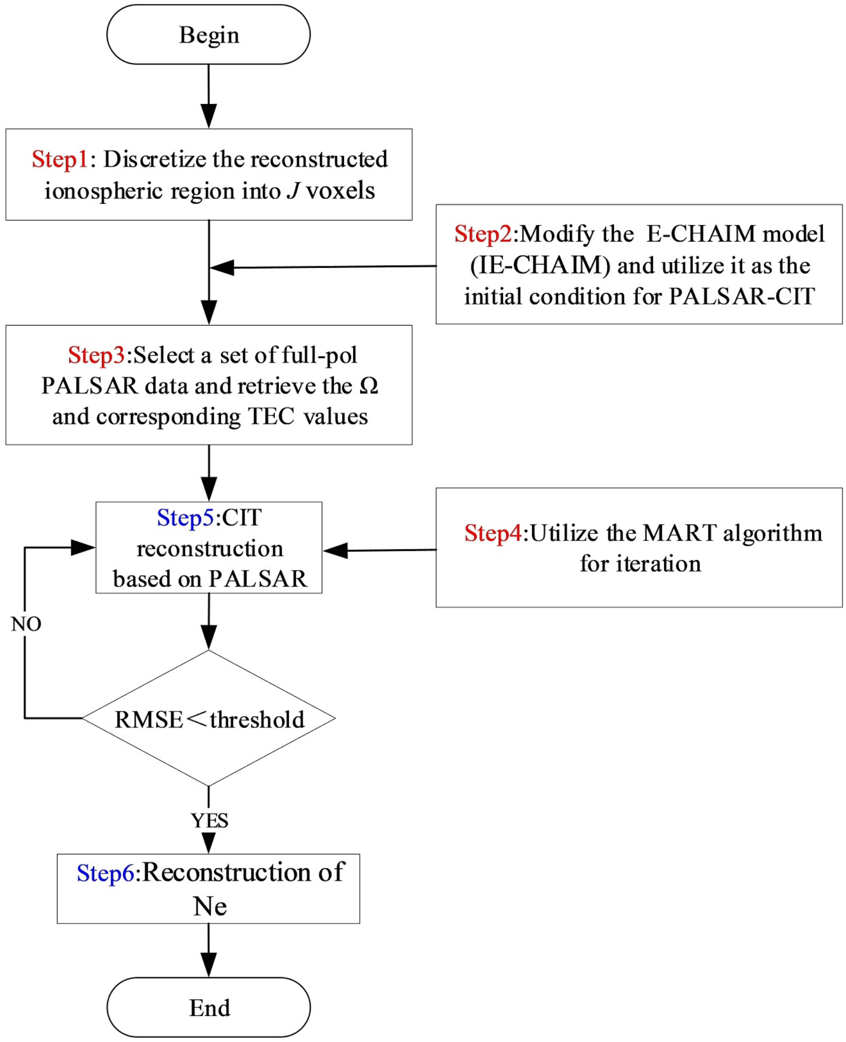

Figure 5. The flowchart of the PALSAR-CIT using the improved E-CHAIM.

discretization of the CIT process, while the latter two steps correspond to voxel-based CIT reconstruction. These aspects will be discussed in detail in Section 3.

3. Performance Verification of PALSAR-CIT With IE-Chaim

Compared with conventional ionospheric sounding techniques, PALSAR exhibits distinct advantages. Its L-band SAR system enables high-resolution observations with all-time and all-weather coverage, independent of ground-based receiving facilities, thereby significantly enhancing spatial observation capabilities. Since low-frequency electromagnetic waves are subject to phase delays and scattering effects while propagating through the ionosphere, the radar echoes acquired by PALSAR during its two-way signal transmission inherently contain integral information along the ionospheric path. This provides a novel and valuable data source for retrieving ionospheric parameters. As a result, this satellite-based radar approach to ionospheric detection demonstrates substantial complementary value and application potential, particularly in high-latitude regions where ionospheric observations are sparse, thus expanding the scientific utility and operational applicability of low-frequency SAR systems in ionospheric monitoring.

However, in the context of constructing voxel-based CIT models using PALSAR data for three-dimensional electron density reconstruction, a major challenge lies in the limited number of available satellites. This constraint leads to insufficient and unevenly distributed PALSAR ray paths causing many discretized ionospheric

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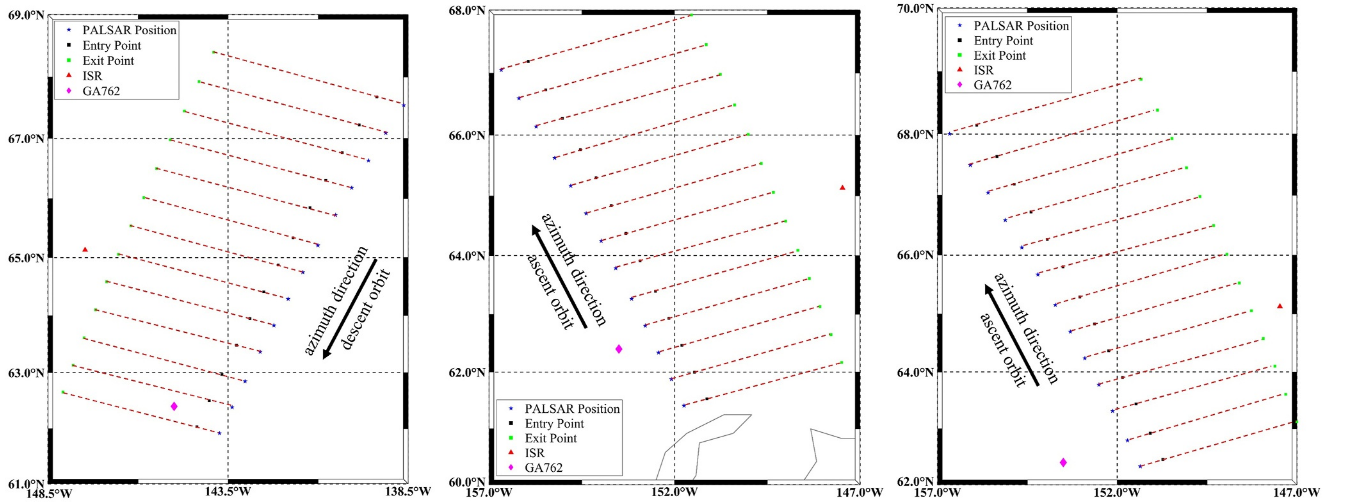

(a) Acquisition at 21:06 UT on August 6, 2010 (b) Acquisition at 07:30 UT on March 19, 2011 (c) Acquisition at 07:28 UT on March 31, 2011

Figure 6. Plan-view maps showing three cases of the spatial distribution of satellite observation points (blue pentagrams), ionospheric ray (red dashed line) entry points at 600 km altitude (black squares), exit points at 100 km altitude (green squares), the location of the ISR station (red triangle), the location of the GA762 ionosonde (purple diamond), and the outlines of the computerized ionospheric tomography reconstruction regions.

voxels to remain uncorrected. Consequently, a significant portion of the voxel electron density values must be assigned via initialization, rendering the reconstruction problem inherently ill-posed. The uncertainty associated with the initial values has long been recognized as a critical factor limiting the accuracy of voxel-based tomographic imaging. In scenarios where increasing the number of ray paths or incorporating additional data sources is not feasible, improving the precision of the initial electron density field becomes the only viable pathway to enhance the quality of tomographic reconstruction. To address this issue, this study proposes an initialization strategy based on a GNSS-driven E-CHAIM model, aiming to provide a more reliable prior electron density distribution and thereby significantly improve the overall accuracy of PALSAR-CIT data (Zheng et al., 2026a) (https://zenodo.org/records/18181621). The algorithm software on which this paper is based is available in Zheng et al. (2026b) (https://zenodo.org/records/18181587).

3.1. Experimental Configuration and Region Selection

Taking into comprehensive consideration the availability of co-temporal PALSAR satellite observations, ISR measurements, and the reconstruction coverage of the E-CHAIM model driven by GNSS, this study selects three cases over Alaska for validation. As shown in Figure 5, this subsection primarily completes the configuration of the necessary parameters involved in the first four steps of the PALSAR-CIT flowchart.

The plan view of the CIT region is shown in Figure 6. For the first case (Figure 6a), the CIT domain spans from 138.5°W to 148.5°W in longitude, 61°N to 69°N in latitude, and extends vertically from 100 to 600 km in altitude. The voxel is configured with spatial resolutions of 5° in longitude, 2° in latitude, and 10 km in altitude. Therefore, a total of 400 voxels represent the IED within the reconstruction region. Thus, the $J = 400$ in step 1. The initial IED (i.e.,) within each voxel is derived from the E-CHAIM or IE-CHAIM model, time-aligned with the PALSAR acquisition at 21:06 UT on 6 August 2010, thereby completing the flowchart of step 2. Within this region, the PALSAR performed an azimuthal scan, capturing a total of 13 PALSAR images with descent orbit (data IDs: ALPSRP241532220 to ALPSRP241532340). By processing the full-pol scattering matrices, the slant TEC values along the corresponding 13 propagation paths are retrieved, with the results being: 7.798 TECU, 7.792 TECU, 7.712 TECU, 7.722 TECU, 7.729 TECU, 7.814 TECU, 7.795 TECU, 7.653 TECU, 7.633 TECU, 7.861 TECU, 7.949 TECU, 7.978 TECU, and 8.072 TECU, respectively. This marks the completion of the parameter configuration for step 3. For step 4, based on the information in the "ceos_leader" and "workreport" files of each PALSAR data set, the coordinates of 13 PALSAR positions and the center point of the scene can be obtained. Based on the above settings, the intercept of each PALSAR ray in the voxels, that is, A_{ij}, can be calculated. In Figure 6a, the 13 PALSAR satellite positions are indicated by blue pentagrams. Additionally, black squares

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denote the points where PALSAR signal rays (red dashed line) intersect the ionosphere at 600 km altitude (entry points), while green squares represent the exit points at 100 km altitude.

Consistent with the voxel size in Figure 6a, the second case (Figure 6b) covers a spatial domain ranging from 147°W to 157°W in longitude, 60°N to 68°N in latitude, and an altitude range from 100 to 600 km. Accordingly, the reconstruction region is discretized into 400 voxels representing the IED. Furthermore, the observation time is at 07:30 UT on 19 March 2011. ECHAIM or IE-CHAIM utilizes the values at the specified time and domain as the initial state for model computation. Similarly, the PALSAR conducted an azimuthal scan over this region, acquiring a total of 13 ascending-orbit images (data IDs: ALPSRP274271250 to ALPSRP274271370). By processing the full-pol scattering matrices, the slant TEC values along the corresponding 13 propagation paths are retrieved. The resulting values are: 1.465 TECU, 1.539 TECU, 1.520 TECU, 1.489 TECU, 1.671 TECU, 1.763 TECU, 1.704 TECU, 1.667 TECU, 1.562 TECU, 1.424 TECU, 1.399 TECU, 1.193 TECU, and 1.089 TECU, respectively. Using the same approach as applied to the PALSAR data in Figure 6a, the corresponding values A_{ij} can also be calculated.

For the third case (Figure 6c), the reconstruction domain is defined between 147°W and 157°W in longitude, 62°N and 70°N in latitude, and spans an altitude range of 100–600 km. Thus, a total of 400 voxels still represent the IED within the entire reconstruction area. The observation time is 07:28 UT on 31 March 2011, which also serves as the input for both E-CHAIM and IE-CHAIM models. An azimuthal scan is performed by the PALSAR within this region, resulting in the acquisition of 13 ascending-orbit images (data IDs: ALPSRP276021270 to ALPSRP276021390). The retrieved slant TEC values along the corresponding propagation paths are 5.323 TECU, 5.267 TECU, 5.226 TECU, 5.195 TECU, 5.228 TECU, 5.426 TECU, 5.749 TECU, 5.828 TECU, 5.938 TECU, 6.232 TECU, 6.443 TECU, 6.619 TECU, and 6.407 TECU, respectively. In the fourth step, after processing the orbital information of PALSAR, the A_{ij} for the third case can be calculated.

Finally, three sets of ISR measurements were retrieved from the Madrigal database for comparative validation. These data sets were obtained from the Poker Flat Incoherent Scatter Radar (PFISR) located near the reconstruction area (147.47°W, 65.13°N) and serve as the ground truth. The acquisition times are closely aligned with the three experimental epochs: 21:08 UT on 6 August 2010, 07:30 UT on 19 March 2011, and 07:28 UT on 31 March 2011.

3.2. CIT Results and Analysis

As shown in Figure 5, after configuring the three data sets required for the experiment, the subsequent 5th and 6th steps are carried out to complete the PALSAR-based tomographic imaging experiment and to validate the proposed algorithm. The iteration is terminated once the RMSE between consecutive iteration results is less than the specified threshold, that is, $3 \times 10^{10} \mathrm{el/cm^3}$. Corresponding to Figure 6, the reconstructed electron density profiles for three epochs, that is, 6 August 2010, 19 March 2011, and 31 March 2011, are compared with the ISR-derived electron density within the reconstruction region (black solid line). Because the grid point closest to the ISR location lies at the edge of the tomographic region, signal rays may not intersect certain altitude levels at the boundary. This may cause the reconstructed electron density at those heights to remain identical to the tomographic initial value. Therefore, instead of directly comparing the ISR measurements with those of the single nearest grid point, we compare them with the average electron density reconstructed from the four points surrounding the ISR location, which is more representative of the actual physical conditions.

Figure 7 presents not only the conventional results obtained from E-CHAIM (blue dashed line) and PALSAR-ECHAIM (blue dash–dotted line) but also the results from the proposed IECHAIM (red dashed line) and PALSAR-IECHAIM (red dash–dotted line). It can be observed that PALSAR-IECHAIM exhibits a closer agreement with the actual electron density distribution (i.e., the ISR results) than the other methods across all three experiments. The corresponding RMSE and AD error metrics for PALSAR-ECHAIM and PALSAR-IECHAIM are summarized in Table 2. Calculations indicate that compared with the three sets of RMSE values reflecting the discrepancies between PALSAR-ECHAIM reconstruction results and the true electron density, the RMSE values of PALSAR-IECHAIM relative to the true values are reduced by 10.1%, 29.94%, and 26.65%, respectively. Meanwhile, the AD values are reduced by 3.83%, 26.88%, and 35.74%, respectively. The contribution of the PALSAR data is reflected in the significant accuracy gain relative to the initial IE-CHAIM.

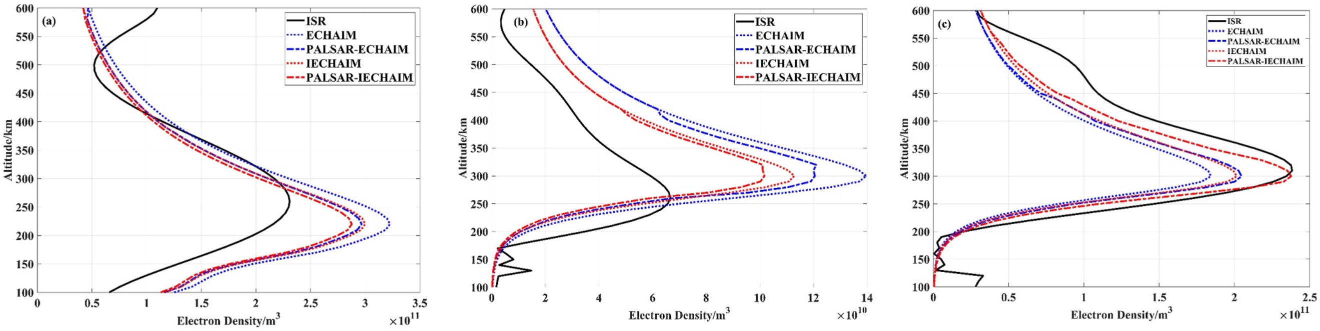

Figure 7. Comparison of electron densities obtained by different methods with ISR measured data at (a) 6 August 2010, (b) 19 March 2011, and (c) 31 March 2011. Unit: el/cm^3.

As shown in Table 2, the RMSE and AD were reduced by up to 13.84%, 14.54%, and 26.97%, 6.87%, 4.75%, and 34.67%, respectively. This confirms that the PALSAR measurements provide a meaningful 'Data Innovation' that drives the reconstruction toward the ISR truth, rather than merely replicating the initial guess.

In addition, it can be observed that, even after the proposed correction, discrepancies between the reconstructed results and the true electron density remain. On the one hand, these discrepancies are partly attributable to the limitations of the ECHAIM model correction, particularly with respect to the peak height and peak electron density, which still require further investigation and refinement. This issue constitutes one of the main directions of our future work. As shown in Figures 7a and 7b, the original ECHAIM results exhibit significant deviations from the ISR measurements in terms of both peak height and peak density. Although these discrepancies are substantially reduced by the proposed algorithm, noticeable differences persist.

On the other hand, the acquisition of PALSAR satellite data is associated with relatively high costs and stringent observational constraints. Consequently, the amount of PALSAR data available in this study remains limited. Under such conditions, although the reconstruction accuracy can be further improved after incorporating PALSAR observations, as indicated in Table 2, the sparsity of the ray paths still leads to an ill-posed reconstruction problem within the region of interest. A greater number of eligible on-orbit low-frequency satellite rays are the key to alleviating this problem, which will be one of the priorities of our future work.

It should be noted that due to the sparse distribution of PALSAR ray paths (e.g., 13 rays in this study), the reconstructed vertical structure, particularly the peak height (hmF2), maintains a degree of dependence on the initial IE-CHAIM model. In voxels not intersected by any signal paths, the electron density values necessarily remain consistent with the initialized state. However, for the voxels traversed by PALSAR rays, the measurements provide a critical 'Observational Increment' or 'Data Innovation' that drives the iterative correction of the electron density numerical distribution. The significant reduction in error metrics—with RMSE and AD decreasing by up to 26.97% and 34.67% relative to the ISR truth, respectively—demonstrates that the PALSAR observations effectively refine the empirical background field. This confirms that while the initial model provides the vertical 'skeleton,' the PALSAR-CIT process delivers high-fidelity numerical updates to the local ionosphere.

Table 2
Accuracy Comparison Between the Two Results and the ISR

	IE-CHAIM		PALSAR-ECHAIM		PALSAR-IECHAIM	
Data	RMSE	AD	RMSE	AD	RMSE	AD
20100806	2.8088e + 10	3.4048e + 10	2.6917e + 10	3.2972e + 10	2.4199e + 10	3.1709e + 10
20110319	1.5089e + 10	2.0679e + 10	1.8406e + 10	2.6935e + 10	1.2895e + 10	1.9696e + 10
20110331	1.6926e + 10	2.9288e + 10	1.6853e + 10	2.9769e + 10	1.2361e + 10	1.9131e + 10

Note. (Unit: El/cm^3).

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4. Conclusions

In this study, the accuracy and practicality of the ECHAIM model are improved through the incorporation of GNSS data, and a PALSAR-IECHAIM–based CIT method is proposed. Validation using the three sets of observational data demonstrates that the significant improvement in reconstruction performance is driven by two main factors. First, the inclusion of PALSAR data provides crucial observational constraints; compared to the IE-CHAIM results, the PALSAR-IECHAIM results reduce the RMSE values by 13.84%, 14.54%, and 26.97%, respectively, while the corresponding AD values are reduced by 6.87%, 4.75%, and 34.67%. Second, the IE-CHAIM model effectively enhances the physical realism of the initial electron density; compared to the PALSAR-ECHAIM results, the PALSAR-IECHAIM results further reduce the RMSE values by 10.1%, 29.94%, and 26.65%, and the AD values by 3.83%, 26.88%, and 35.74%, indicating a significant improvement in reconstruction performance. Conventional CIT generally relies on GNSS observations and thus requires the deployment of ground-based receivers. However, in polar regions, large areas remain poorly covered due to the scarcity of GNSS infrastructure. The hybrid strategy of integrating the GNSS-TEC-driven IE-CHAIM model with the PALSAR-CIT method proposed in this paper demonstrates that without relying on the GNSS system, the two-way propagation characteristics of spaceborne low-frequency SAR signals can be exploited to reconstruct the spatial distribution of IED without the need for ground receivers, thereby exploring and evaluating the feasibility of independent PALSAR-based tomographic imaging.

While a single PALSAR mission cannot fully resolve all vertical structures (like hmF2) independently, it establishes a robust framework for spaceborne ionospheric sounding in remote regions. The fidelity of this technique lies in its ability to refine empirical models with real-time satellite observations. Future missions involving multi-satellite constellations will likely further mitigate the sparsity of ray paths and improve tomographic resolution.

The present work provides a robust methodological baseline for future ionospheric sounding that integrates spaceborne low-frequency full-polarimetric SAR systems with the IE-CHAIM model. Beyond a supporting role, this framework provides a standardized protocol for integrating multi-source data to ensure the fidelity of trans-ionospheric radar imaging. Given its generalizable nature, this approach offers significant potential for upcoming L-/P-band missions, such as BIOMASS SAR, PALSAR-3, and LT-1A/B, providing unprecedented opportunities for high-fidelity ionospheric observation (Ji et al., 2025).

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The GNSS, ISR and PALSAR data can be available at Zheng et al. (2026a, 2026b).

Acknowledgments

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